

Champion-chain signal: a pre-tournament filter for real title contenders

A reproducible backtest of the Champion-Chain Signal (CCS), plus a practical 1998 to 2026 pre-tournament lens for identifying recognizable contenders that deserve a champion-probability downgrade.

Executive Summary

- **CCS is a candidate-pool filter, not a champion picker.** In the 1986-2022 modern knockout era, CCS retained only 30.1% of team-tournaments but covered 9/10 (90.0%) of all champions; after excluding the one no-prior-history case, coverage was 9/9 (100.0%).
- **The result is not a random one-third screen.** A random candidate pool with each year's same size would expect only 3.0 champion hits out of 10; the probability of randomly reaching at least 9 hits is 0.012%.
- **A stronger simulation still supports the signal, but with humility.** After preserving the mix of traditional title contenders and ordinary teams, a label-permutation control expects 4.0 hits out of 6 evaluable champions; reaching 6/6 is a 8.2% event.
- **The most useful reader experience is the pre-tournament downgrade list.** From 1998 onward, a long list of recognizable contenders were highly ranked but non-CCS at kickoff. Many still advanced, including finalists, but none won in the modern sample.
- **The mechanism is partly strength, but more specific than strength.** CCS overlaps with elite teams, yet it asks a narrower question: has this team recently been on, or directly removed from, the champion path?

9/9
(100.0%)

9/10
(90.0%)

30.1%
Modern candidate-
pool share

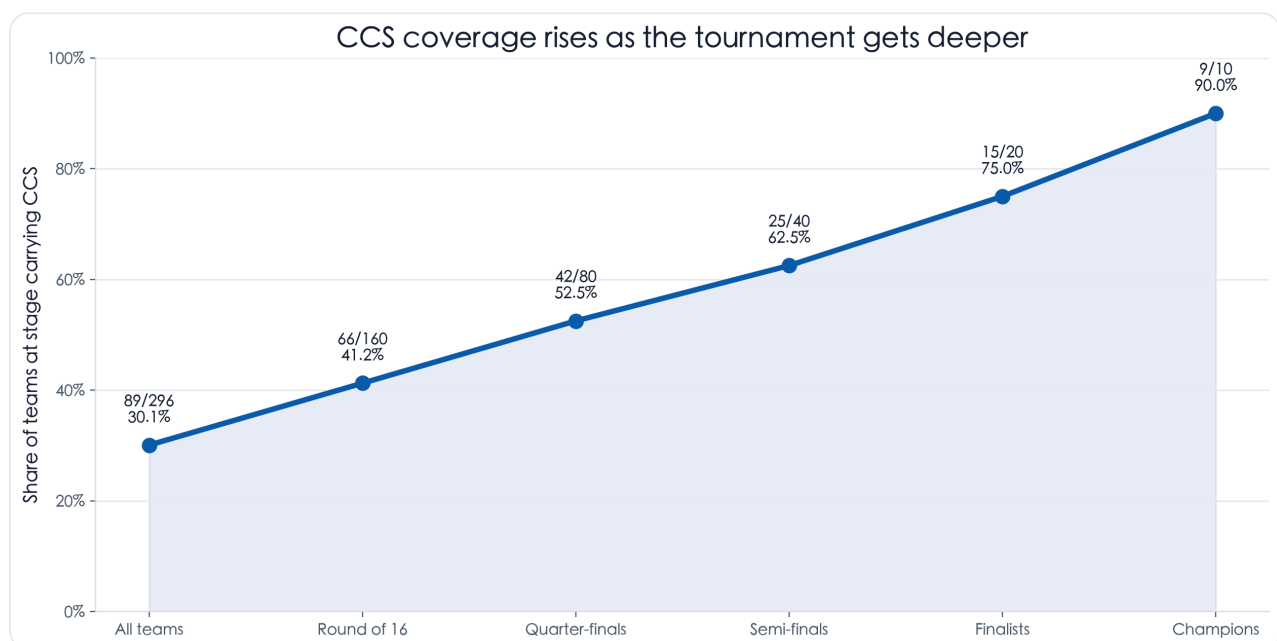
0.012%
Random same-size
pool reaches $\geq 9/10$

Evaluable modern
champions covered

All modern
champions covered

1. CCS gets more concentrated as the tournament gets serious

The primary backtest pattern is monotonic. CCS teams are about three-tenths of the field, but their share rises at every deeper stage: round of 16, quarter-finals, semi-finals, final, and champion. This makes the signal more useful as a championship filter than as a generic knockout-stage prediction tool.



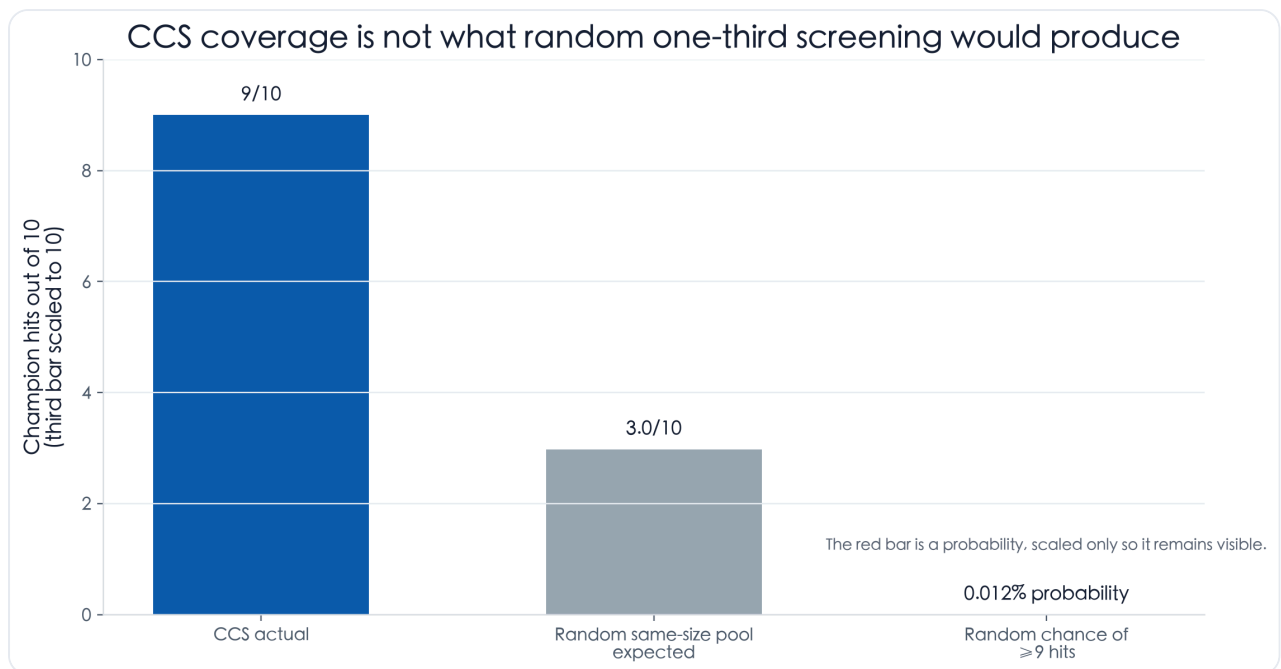
Modern era is 1986–2022. Champion coverage is 9/10 on the all-champion denominator and 9/9 after removing France 1998, which had no prior-two-World-Cup finals history.

2. Method and simulation: stronger than random, not proof by itself

The method has two layers. First, CCS is defined before the tournament from the prior two World Cups only: a team qualifies if it recently won the World Cup or was knocked out by a later finalist. Second, the report tests whether that pre-tournament label covers champions better than no-information and strong-team controls.

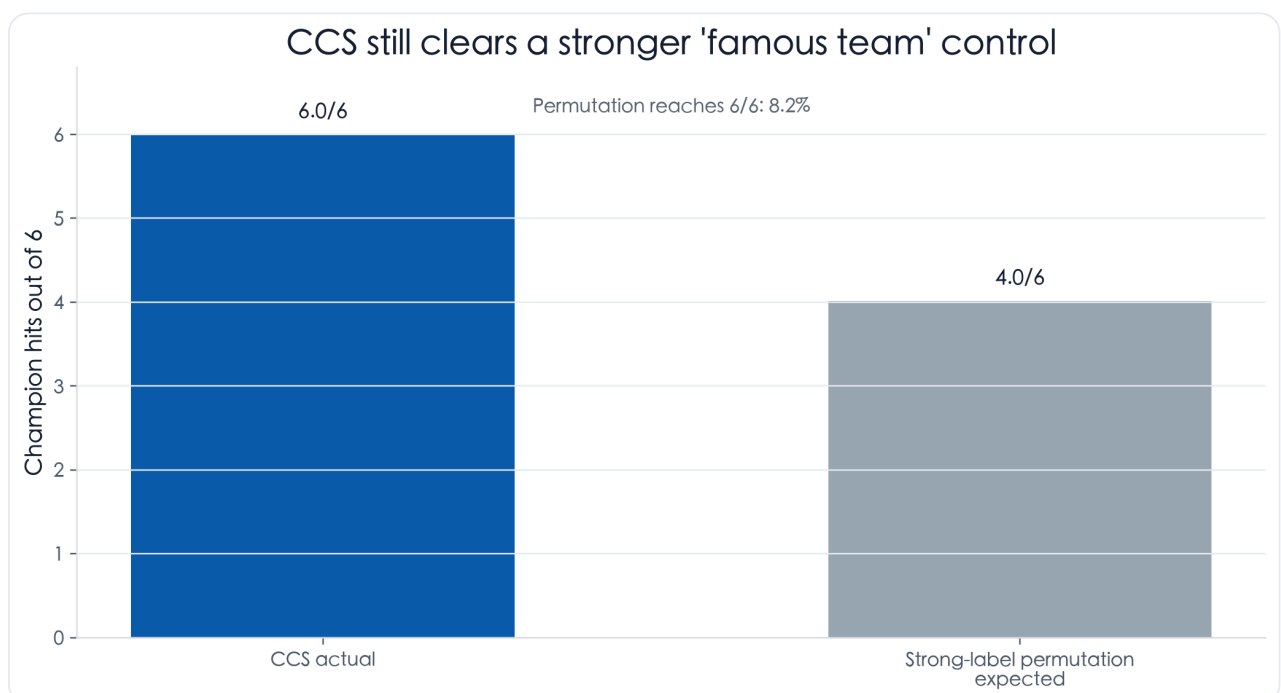
A same-size random pool is the first hurdle. If 1998 has 11 CCS candidates out of 32 teams, the random benchmark also randomly selects 11 of 32. Repeating that across

the ten modern tournaments yields an expected 3.0 champion hits, not nine. Hitting at least nine by chance is a 0.012% event.



The benchmark preserves each year's candidate-pool size, so it tests information content rather than simply rewarding CCS for selecting more teams.

The stronger control asks whether CCS is merely a famous-team label. For each tournament from 1998 to 2022, we preserve the number of CCS labels inside the traditional title-contender set and outside it, then randomly permute the labels inside those two groups. Excluding the explicit France 1998 no-prior-history exception, CCS hit 6/6 evaluable champions; the strong-label permutation expects 4.0/6, and reaches 6/6 with probability 8.2%.

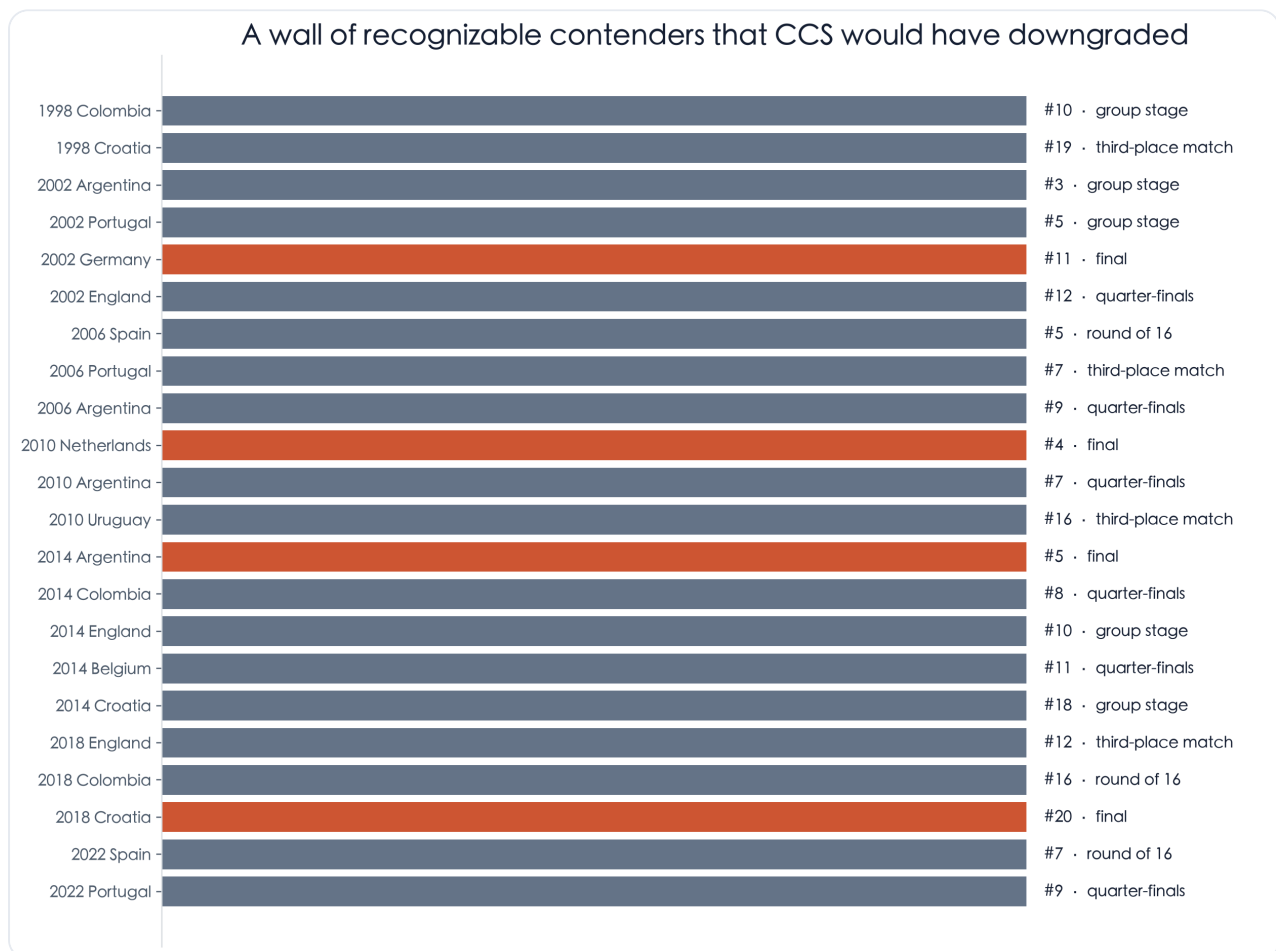


This is deliberately not a FIFA-ranking simulation. It controls for the broader fact that CCS often overlaps with obvious football powers, then asks whether the specific champion-chain label still carries information.

Interpretation discipline: these simulations support CCS as a credible screening heuristic. They do not prove it beats Elo, betting odds, or a full multivariate model. The next bar is incremental value versus those stronger baselines.

3. The pre-tournament experience: recognizable favorites CCS would downgrade

This is the most intuitive way to use the method. Before kickoff, a team can be highly ranked, historically recognizable, and still lack a recent champion-chain connection. The main exhibit is curated from a Top-20 non-CCS audit pool to show the teams a modern audience would naturally treat as title-relevant: Argentina, Germany, England, Spain, Portugal, Netherlands, Belgium, Colombia, Croatia, and Uruguay.



Curated from qualified teams that were FIFA Top 20 and non-CCS at kickoff. The full mechanical Top-20 audit table is retained in `data/derived/favorite_traps.csv`; the curated list is retained in `data/derived/favorite_trap_powerhouses.csv`.

Year	Team	Code	FIFA rank	Final outcome	Ranking date
1998	Colombia	COL	#10	group stage	1998-05-20
1998	Croatia	HRV	#19	third-place match	1998-05-20
2002	Argentina	ARG	#3	group stage	2002-05-15
2002	Portugal	PRT	#5	group stage	2002-05-15
2002	Germany	DEU	#11	final	2002-05-15
2002	England	ENG	#12	quarter-finals	2002-05-15
2006	Spain	ESP	#5	round of 16	2006-05-17
2006	Portugal	PRT	#7	third-place match	2006-05-17
2006	Argentina	ARG	#9	quarter-finals	2006-05-17
2010	Netherlands	NLD	#4	final	2010-05-26
2010	Argentina	ARG	#7	quarter-finals	2010-05-26
2010	Uruguay	URY	#16	third-place match	2010-05-26
2014	Argentina	ARG	#5	final	2014-06-05
2014	Colombia	COL	#8	quarter-finals	2014-06-05
2014	England	ENG	#10	group stage	2014-06-05
2014	Belgium	BEL	#11	quarter-finals	2014-06-05
2014	Croatia	HRV	#18	group stage	2014-06-05
2018	England	ENG	#12	third-place match	2018-06-07
2018	Colombia	COL	#16	round of 16	2018-06-07
2018	Croatia	HRV	#20	final	2018-06-07
2022	Spain	ESP	#7	round of 16	2022-10-06
2022	Portugal	PRT	#9	quarter-finals	2022-10-06

The pattern is useful but not absolute. Non-CCS strong teams can go deep: 2002 Germany, 2010 Netherlands, and 2014 Argentina reached finals. The historical point is narrower and stronger: in the modern sample, the champion almost always came from the CCS side of the field.

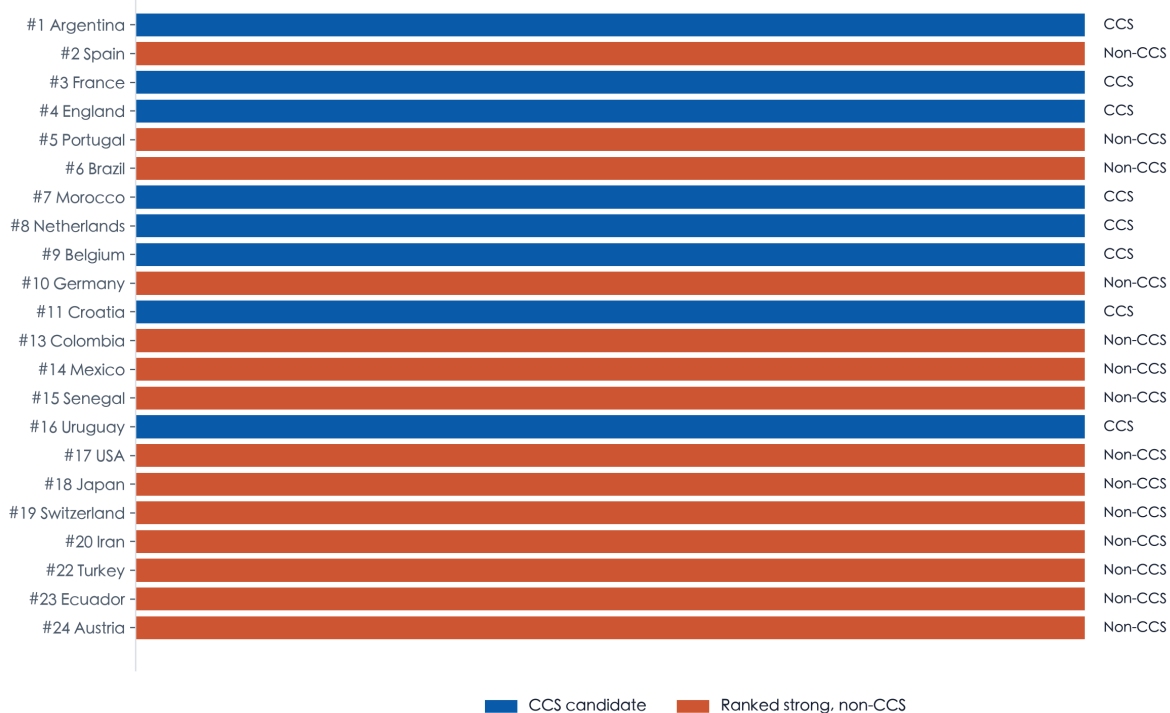
4. 2026 application: separate rank strength from champion-chain strength

The 2026 view is a live-use case, not a backtest result. The ranking snapshot is frozen at FIFA's June 11, 2026 official ranking and the qualified-team list is from FIFA's 2026 season endpoint. The headline application is clear: Spain, Portugal, Brazil, Germany, and Colombia are rank-strong, reputation-strong, but non-CCS before kickoff.

fifa_rank	Team	Code	FIFA points	Downgrade reason
#2	Spain	ESP	1874.71	Rank/reputation strong, but no CCS source from 2018/2022
#5	Portugal	POR	1767.85	Rank/reputation strong, but no CCS source from 2018/2022
#6	Brazil	BRA	1765.86	Rank/reputation strong, but no CCS source from 2018/2022
#10	Germany	GER	1735.77	Rank/reputation strong, but no CCS source from 2018/2022
#13	Colombia	COL	1698.35	Rank/reputation strong, but no CCS source from 2018/2022

The broader watchlist below keeps the full top-ranked context visible, so the downgrade call is not hidden inside a hand-picked list.

2026 pre-tournament watchlist: which contenders lack CCS support



Top 24 ranked qualified teams in the 2026 field. Blue teams have CCS support from 2018 or 2022; red teams are rank-strong but non-CCS.

fifa_rank	Team	Code	CCS status	CCS source World Cup
#1	Argentina	ARG	CCS candidate	2018;2022
#2	Spain	ESP	Non-CCS downgrade	none
#3	France	FRA	CCS candidate	2018;2022
#4	England	ENG	CCS candidate	2018;2022
#5	Portugal	POR	Non-CCS downgrade	none
#6	Brazil	BRA	Non-CCS downgrade	none
#7	Morocco	MAR	CCS candidate	2022
#8	Netherlands	NED	CCS candidate	2022
#9	Belgium	BEL	CCS candidate	2018
#10	Germany	GER	Non-CCS downgrade	none
#11	Croatia	CRO	CCS candidate	2018;2022
#13	Colombia	COL	Non-CCS downgrade	none
#14	Mexico	MEX	Non-CCS downgrade	none

fifa_rank	Team	Code	CCS status	CCS source World Cup
#15	Senegal	SEN	Non-CCS downgrade	none
#16	Uruguay	URU	CCS candidate	2018

5. Why this happens: strength matters, but path matters too

CCS partly captures strength, and that should be acknowledged. Champions, finalists, and teams beaten by finalists are usually strong teams. A signal built from those events will naturally overlap with FIFA ranking, odds, and historical reputation.

But CCS is not simply 'teams that often qualify' or 'teams that are highly ranked.'

It requires a specific recent relationship to the champion path: either winning the World Cup or being knocked out by a finalist. A team can be ranked highly, qualify regularly, or have reached a quarter-final and still be non-CCS if it did not touch that path.

The proposed mechanism is tournament-cycle validation. Recent champion-chain contact is a proxy for having already met World Cup knockout intensity against finalist-level opposition. It is not causal proof; it is a compact historical state variable that seems especially relevant for champions rather than finalists.

Recommended Next Steps

1. **Use CCS as the first-layer title filter.** Start the champion pool with CCS candidates, then require extra evidence to re-admit non-CCS teams.
2. **Add a stronger model benchmark.** Compare CCS against Elo, odds-implied probabilities, FIFA ranking, and a combined model to test incremental value.
3. **Track 2026 without moving the goalposts.** Freeze the pre-tournament CCS/ranking table and evaluate it only after the tournament is complete.

Further Questions

- Does CCS add predictive lift after controlling for Elo or betting odds?
- Is a one-, two-, or three-tournament lookback optimal once tested out of sample?

- Should hosts, major squad-cycle upgrades, and no-prior-history teams receive a separate override flag rather than being treated as ordinary non-CCS teams?
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Caveats And Assumptions

- The modern champion sample has only 10 observations. The 100% evaluable number is 9/9, not a stable law.
 - France 1998 is treated as a no-prior-history exception because it missed the prior two World Cup finals tournaments.
 - FIFA ranking is used as a public strong-team proxy; odds and Elo would be better next-step baselines.
 - 2026 outcomes are not used in the 2026 watchlist. The ranking date is frozen at June 11, 2026.
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Generated by scripts/build_report.py. Sources: Fjelstul World Cup Database; FIFA API ranking schedules, rankings, and 2026 qualified-team endpoint. Original corrected CCS report is preserved in reports/pdf/original_ccs_report_corrected.pdf.